

University's Study Space Booking Systems

System Design and Analysis

Maastricht University

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1. Introduction

Maastricht University has about 22.500 students (Maastricht University) enrolled in their different bachelor's and master's programs. Not all students have the luxury of studying at their home, and some students simply prefer studying at the facilities provided by the university. To address this, the university offers a variety of study locations, namely the inner city library, Paul-Henri Spaak, and Tapijn Loods V Learning spaces, in which students can work with or without reservation. Additionally, students can make room reservations within the different faculty buildings.

This paper aims to critically analyze the systems that the university currently has in place for reserving study spaces, namely the individual and the group booking systems. Users can find the corresponding applications on the university's website. Using context diagrams and level-0 Data Flow Diagrams (DFD), we will analyze the structure of the current systems. A small survey that we conducted among students will further show the weakness of the current method. With this analysis, we will propose ideas for a more efficient and effective system, with the help of redesigned user interfaces, updated context diagrams, Data Flow Diagrams (DFDs), Business Process Model and Notation (BPMN) and Entity-Relationship (ER) diagrams.

2. Current Personal Booking System

2.1 Context Diagram

A context diagram is an overview of an organizational system that shows the boundaries, external entities that interact with the system, and any major information flows between entities and the system (Valacich & George, 2022). Figure 1 illustrates the application of context diagram on the current booking system for individuals in Maastricht University.

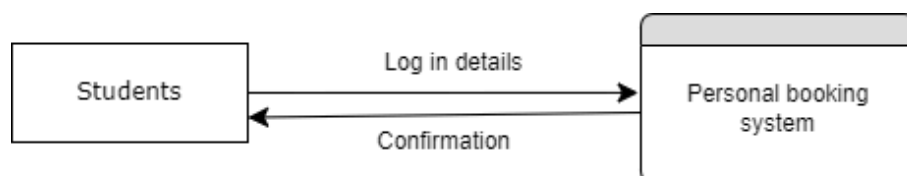


Figure 1: Context diagram of the current personal booking system

2.2 Level-0 diagram

The level-0 diagram represents a system's major processes: data flows and data stores at a high level of detail (Valacichi & George, 2022). This allows us to have a clear understanding of the steps the user has to take to go through the actual process of booking a study spot.

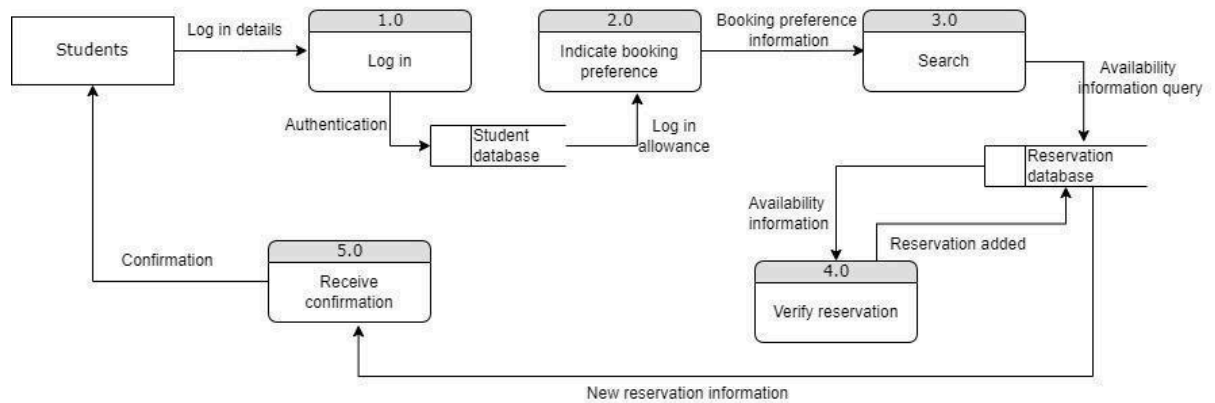


Figure 2: Level-0 DFD of the current personal booking system

1.0 Log in

As depicted in Figure 2, the process starts with a student, or the user, logging in with their login details; this brings us to the first process 1.0 Login. The information the user uses to log in then gets authenticated and cross referenced with the information in the student database. When the information that is being used by the student is correct, the system will allow the student to log in to the system. With this step, we make the assumption that every student has an account and knows their login details. Additionally, we had to simplify the login process as we do not have access to all the information and database structures that are normally behind the logging-in process.

2.0 Indicate booking preference

When the student is logged in to the system, they can indicate their booking preference. They can indicate any preferences that they might have for their booking. Once they have completed this, they can start the search.

3.0 Search

In process three, the actual search for all options that match the search requirements takes place. This is done by running an availability information query on the data in the room timetable database. We make the assumption that all available study spots that students can reserve in the Library have been added to this database.

4.0 Make Reservation

Once the query has been running the data can be used to make a reservation. The system will, after the reservation has been made, send the information on the booking back to the Reservation Database. The system checks if the availability is correct and will send out a message to confirm the booking.

5.0 Receive Confirmation

When the booking is confirmed and the database information is updated a message will be sent out to the student to confirm their booking.

3. Current Group Booking System

The second application that the university has in place to make study reservations is for room bookings. This application runs separately from the library's individual study spot booking system.

3.1 Context Diagram

Similar to the context diagram we saw with the individual study spot booking system, the context diagram for the room booking system is also fairly straightforward, which can be observed from figure 3. A student logs into the website with the login details and gets access to the group room booking system. In the system, the student can make the reservation (not shown in a context diagram), and the system then sends a signal back to the students confirming their reservation.

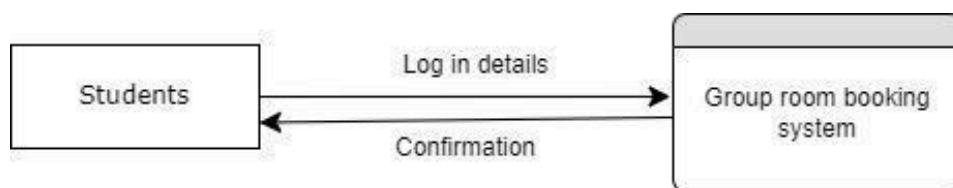


Figure 3: Context diagram of current group booking system

3.2 Level-0 DFD

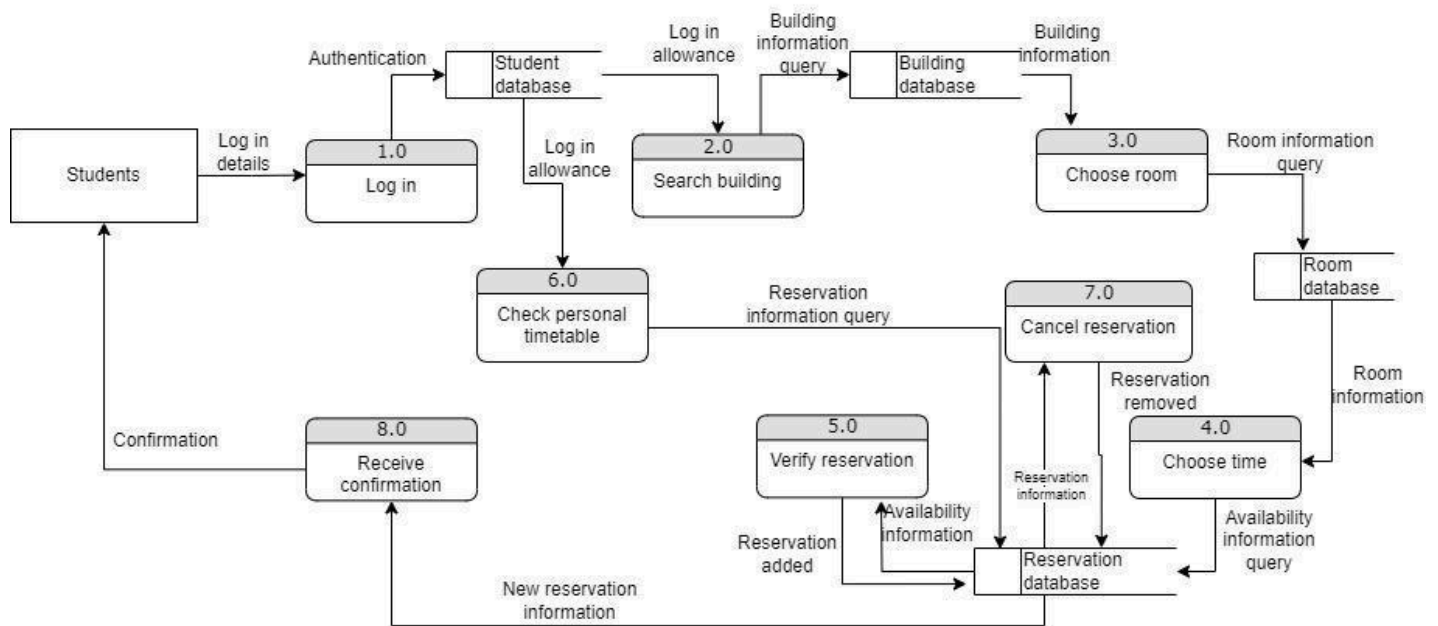


Figure 4: Level-0 DFD of the current group booking system

1.0 Log in

Also starting with the students, the process begins as they log in with their login details, which brings us to the first process, 1.0 Log in in figure 4. The information the user uses to log in then gets authenticated and cross referenced with the information in the student database. When the information being used by the student is correct, the system will then allow the student to log in to the system. With this step, we make the assumption that every student has an account and knows their login details. Additionally, we had to simplify the login process as we do not have access to all the information and database structures that are normally behind the logging-in process.

2.0 Search Building

Once the user is logged into the system, they are presented with the option to make a room booking or make a room booking in the library building. Depending on their choice, the system will find all available buildings using the building information query from the building database. The information found in the query will then be presented to them in the following process.

3.0 Choose Room

In this step, the user will be presented all the rooms that are bookable. Here they can pick what rooms they would like to select based on the preferences they have like room capacity or the time availability of the room. Once the user has added this information, the room information query will be run through the room database.

4.0 Choose Time

Based on the information that is presented with the room information query the student has to pick the time that they would like to book the available room. With the information availability query the system will double check if the room is available during the selected timeframe in the reservation database.

5.0 Verify Reservation

When the room is available the booking is verified and the reservation is added to the reservation database.

8.0 Receive information

When the reservation is added to the database, a message will be sent out to the student to confirm their booking.

6.0 Check Timetable

Alternatively, students also have the option to check their timetable to see an overview of all the bookings that are in place.

7.0 Cancel Reservation

If a student wishes to do so, they can cancel their reservation. The reservation will be removed, and the reservation database will be updated.

4. Problem Analysis

Based on the diagrams and the interface we are already able to highlight some of the main problems in the current booking systems. However, we found it also important to do a short survey in which we asked the students how they used the platforms and if they had any notes regarding the booking system. In total, 20 people filled out our questionnaire; this is not a large group of participants; however, it does give us a feeling of how students judge the systems that the university has in place. Of the total 20 students, 75% of them answered that they do make reservations for individual study spots at the library, whereas only 50% answered that they make reservations for rooms in the university. Looking at the reason why students do not make reservations for rooms, we found out that this is because either they do not know this is possible, or they find the booking system to work with complicated. Students additionally commented that they find the system to be outdated, and unusable and find the

libraries to be overcrowded. Most of the people making reservations do so on their computer (55%) and the rest of the students make their reservations on the phone.

Looking at the system that is currently in place, we believe that the main problem is the complexity. We believe that having two different systems in place to make reservations makes the process of making a reservation unnecessarily complex and not user-friendly. This results in students not knowing that they are able to make room reservations. Additionally, we believe that there are some things currently missing in the filter for both the individual booking systems as the group booking system. It is currently hard to find free spots that are 'available now' and would like to add a filter in the system to make this easier. We also believe that it is hard to figure out the location of the rooms in the group booking system and would like to make this easier with additional filters as well.

5. Proposed New System

5.1 Context Diagram

On the context level, there is also not much difference compared to the current system. Students are still the only entity interacting with the system. By sending the log in details, they get to receive the confirmation of their reservation through email.

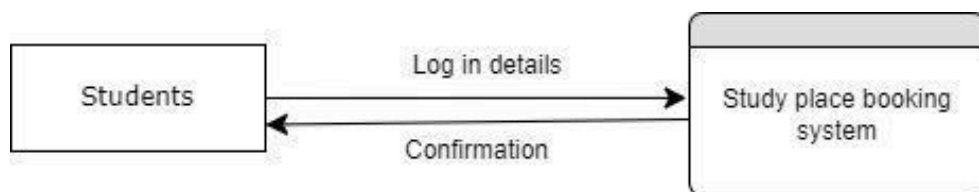


Figure 5: Context diagram of the proposed new system

5.2 Level-0 DFD

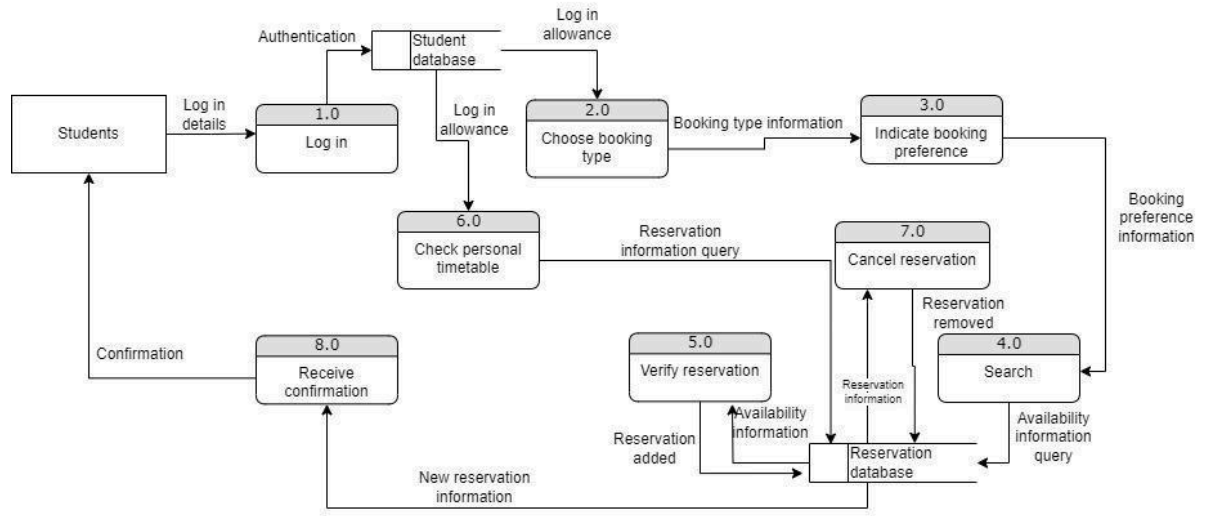


Figure 6: Level-0 DFD of the proposed new system

To fix the complexity of having the 2 separated booking systems on 2 different websites, we decided to merge the two websites via the addition of the process to choose booking type in the very beginning. The inefficient processes are also removed to make room for 1 simple process for booking preference. The cancel reservation feature from the current group booking system is also kept.

1.0 Log in

Being similar to the current systems, “log in” process requires user to send their log in details including username and password, so they would send the authentication to the student database, which grants the student with the log in allowance

2.0 Choose booking type

This is the most important update of the system. To simplify the process for student, they can indicate straight away if they would like to make a group room booking or a personal study spot. The proposed interface corresponding to this update can be found in Appendix A. By receiving log in allowance, the student can now click on their preferred book type, sending information to the system.

3.0 Indicate booking preference

To simplify the current version which includes “choose building”, then “choose room”, and “choose time availability” one by one on different pages, in this updated version,

students can indicate all their booking preferences in the same page. The booking preference details will be clarified with the help of the BPMN model in the next section.

4.0 Search

Having received the booking preference information, the system now allows students to just click on “search” and it will thereby send the availability information query to the Reservation database. The database will now show the availability information to the student through the website.

5.0 Verify reservation

As all the information is now insight, the student can easily verify their reservation, which would send the information of a new reservation added to the Reservation database.

6.0 Check personal timetable

After logging in to their account on the website, the student can also have the option to click on the button to check their personal timetable if they wish to cancel their previously made reservation. We believe that the function to cancel reservations like in the group booking system is necessary, which is why we decided to include it instead of not having the option to cancel like the current personal one. Checking the personal timetable will send a Reservation information query to the Reservation database, which would send back the reservation information to the user for the next step

7.0 Cancel reservation

To accomplish their demand to cancel a reservation, users can just click to cancel, and the information of an old reservation is removed from the Reservation database.

8.0 Receive confirmation

Whether the student chooses to make or cancel the reservation on the website, the Reservation database will send the new reservation information to the system, which would generate a confirmation email to the student

5.3 BPMN

As the students are the users of the system, it can be observed that most of the activities take place in the Students lane in the model. Having started the event with the aforementioned log in step, students then face an XOR gateway where they can either choose

to “Choose booking type” to later on proceed to make a reservation, or they can “Check personal timetable” if they wish to cancel their reservation. If they choose the latter, the process would happen exactly as described in the section 5.2, where they just proceed directly to “Cancel reservation” activity.

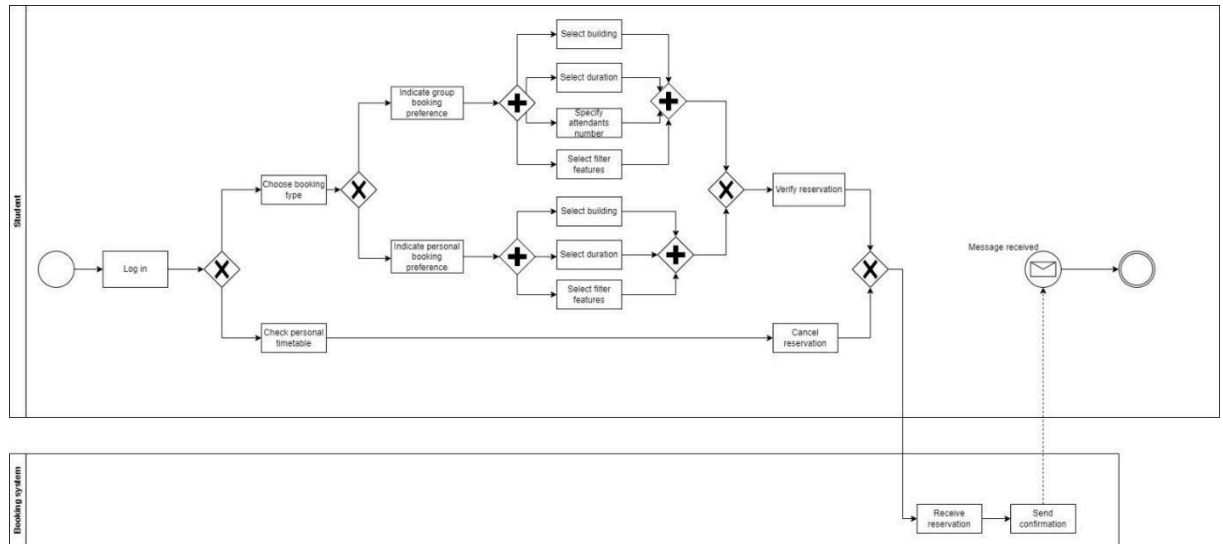


Figure 7: BPMN of the proposed new system

On the other hand, if a user decides to choose booking type, they would face a choice of making group room booking or a personal booking, represented by an exclusive gateway. In the case of choosing to book a group room, they will face an AND gateway where they simultaneously must make the choices of building, duration of booking, and number of attendants in the booking. Additionally, they can also select the features of the room that they would like in the features filter, which includes Wheelchair Accessibility, Screen, Movable Furniture, PC/Beamer, Hybrid Meeting, and Power Availability. Those features will be shown as check-boxes for users to tick in if they specifically demand the feature. The user interface of this stage can be found in Appendix B.

If the user chooses personal booking, they would also be required to specify the building, duration of booking, and also the filter features, but this time with just 2 features, which are Wheelchair Accessibility, and Power Availability. This is because several features targeted for room facilities are no more applicable to a personal study spot. The user interface corresponding to this can be found in Appendix C. After the choices, the gateway closes, leading the user to verify their reservation, which is the same step for users regardless of them having chosen the group booking or the personal booking initially.

In both cases where users decides to either make reservation or cancel reservation after logging in, the new reservation information will be received by Maastricht University booking

system. The system will proceed to send a confirmation email to the student, which ends the event

5.4 ER Diagram

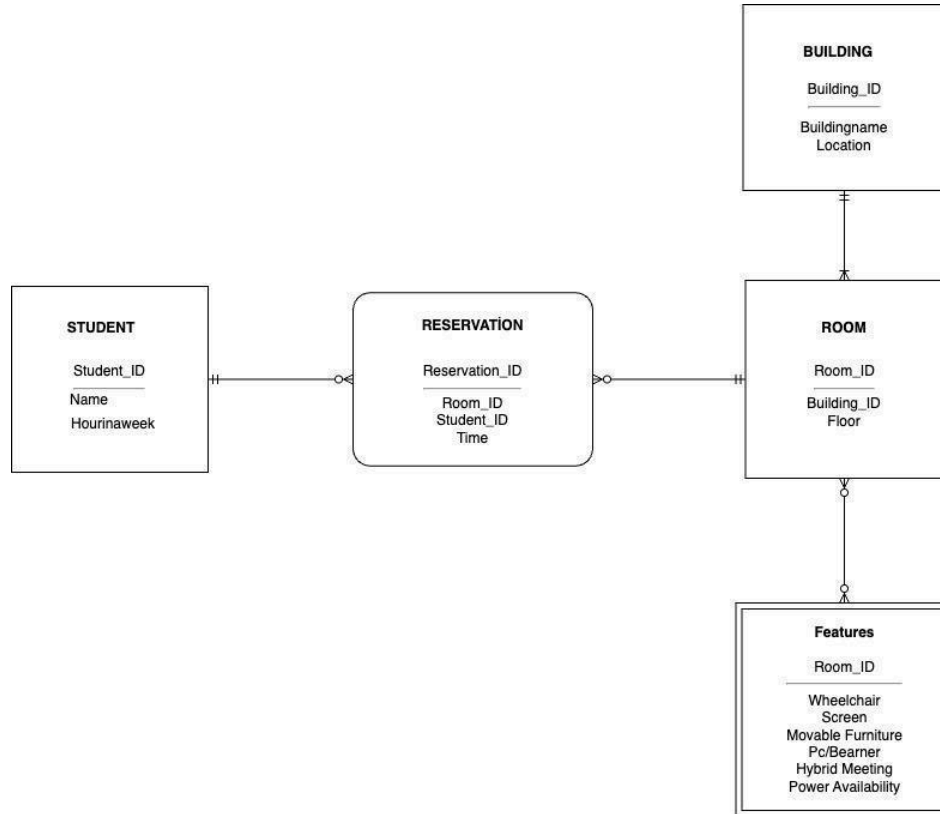


Figure 8: ER Diagram of the proposed new system

Demonstrating the data aspect of the proposed new system, the ER diagram in figure 8 consists of 5 entities, which are Room, Student, Building, Reservation, and Features, each with their own attributes and relationships. 4 databases corresponding to 4 entities of Student, Building, Reservation and Features can be found in Appendix D.

Student: Identified by Student_ID (PK), attributes include name and the number of hours booked for a room in a week.

Room: Identified by Room_ID (PK), associated with Building_ID (FK), and attributes include floor number.

Building: Identified by Building_ID (PK), with attributes such as building name and location.

Reservation: Associated entity identified by Reservation_ID (PK), associated with both Student_ID and Room_ID (FK), with time as another attribute.

Features: A weak entity associated with Room_ID, describing amenities like Wheelchair Accessibility, Screen, Movable Furniture, PC/Beamer, Hybrid Meeting, and Power Availability.

Relationships:

Student to Reservation: One student can make multiple reservations, but each reservation must have one student associated with it.

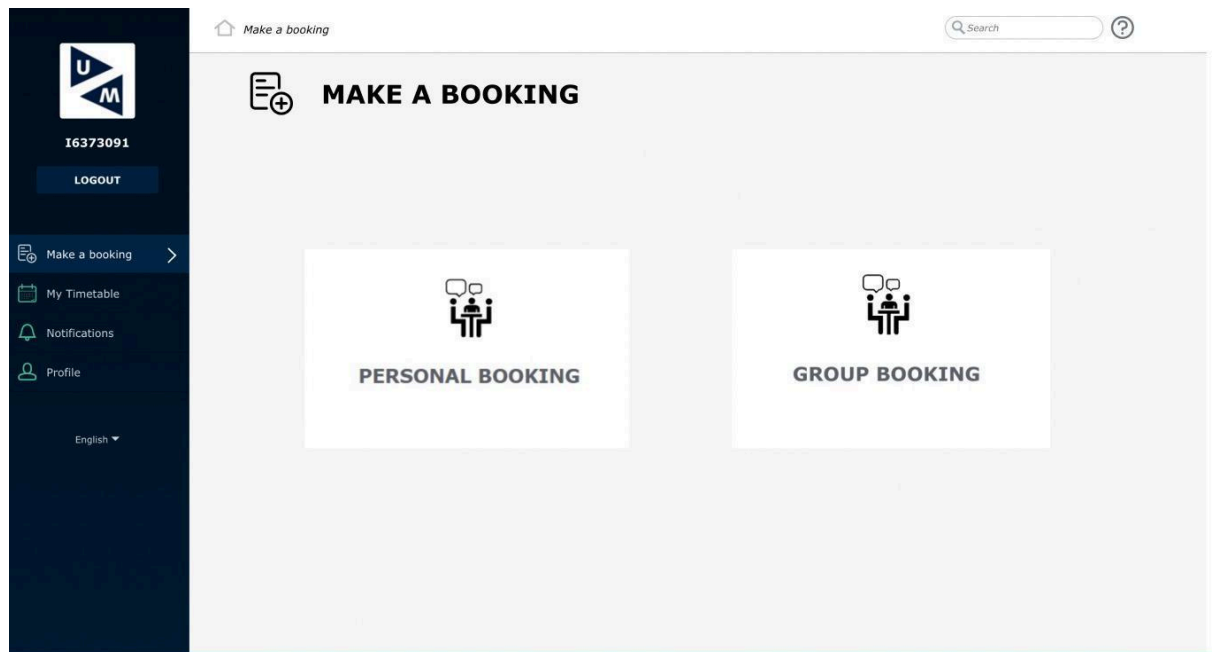
Room to Reservation: One room can have multiple reservations, but each reservation must be for exactly one room.

Room to Features: Rooms can have multiple features, and features can be associated with multiple rooms.

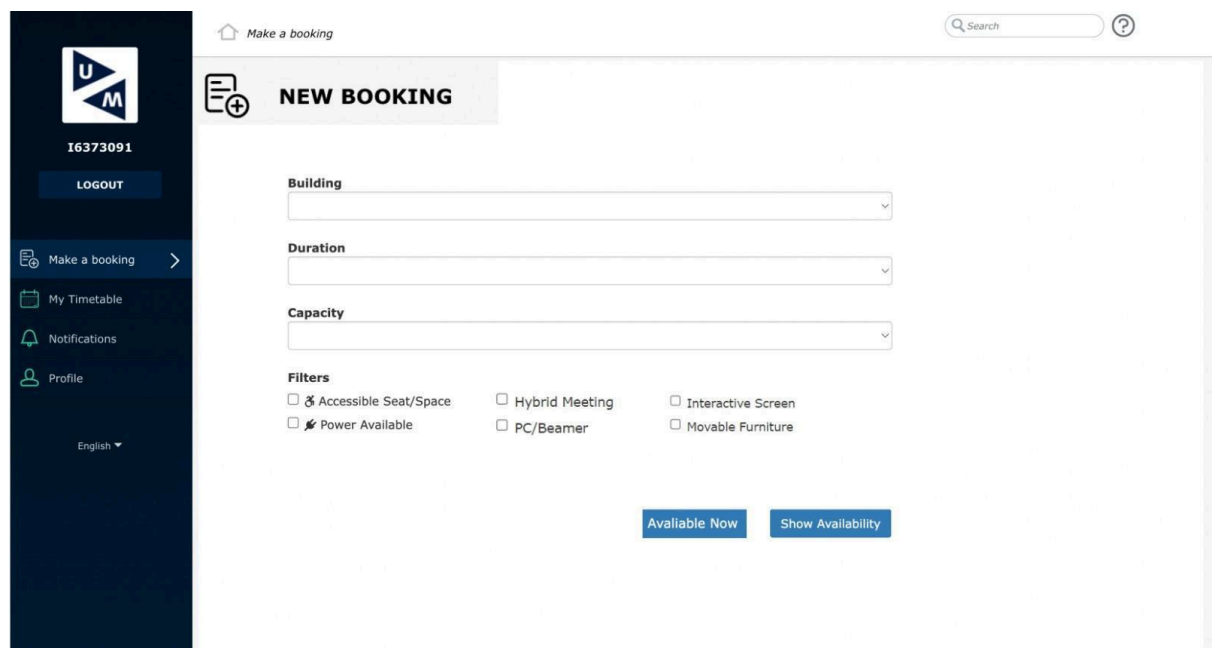
Room to Building: Each room belongs to one building, ensuring each building has at least one associated room.

Appendix

Appendix A: User interface for stage 2.0 Choose booking type



Appendix B: User interface for students indicating group booking preference



Appendix C: User interface for students indicating personal booking preference

The screenshot displays a web application interface for making a new booking. On the left is a dark blue sidebar with the University of Manchester logo, the user ID 'I6373091', a 'LOGOUT' button, and a menu with options: 'Make a booking' (selected), 'My Timetable', 'Notifications', and 'Profile'. At the bottom of the sidebar is a language selector set to 'English'. The main content area has a header with a home icon, the text 'Make a booking', a search bar, and a help icon. Below the header is a 'NEW BOOKING' section with a plus icon. The form includes two dropdown menus for 'Building' and 'Duration'. Under the 'Filters' section, there are two checkboxes: 'Accessible Seat/Space' and 'Power Available', both of which are currently unchecked. At the bottom right of the form are two buttons: 'Available Now' and 'Show Availability'.

U
M

I6373091

LOGOUT

Make a booking >

My Timetable

Notifications

Profile

English ▾

Make a booking

Search ?

NEW BOOKING

Building

Duration

Filters

☐ Accessible Seat/Space

☐ Power Available

Available Now

Show Availability

Appendix D: Relational Database Tables

Student Table:

Student_ID	Name	Hourinaweek
1001	John Doe	10
1002	Jane Smith	15
1003	Alice Lee	8
1004	Bob Brown	12

Features Table:

Room_ID	Wheelchair	Screen	Movable Furniture	PC/Beamer	Hybrid Meeting	Power Availability
101	Yes	No	Yes	Yes	No	Yes
102	No	Yes	Yes	No	No	Yes
201	No	No	No	Yes	Yes	Yes
302	Yes	Yes	No	Yes	No	Yes

Building Table:

Building_ID	Buildingname	Location
1	Campus Venlo	Location A
2	FASoS Buildings	Location B
3	FHML Buildings	Location C
4	SBE Buildings	Location D

Reservation Table:

Reservation_ID	Time	Room_ID	Student_ID
1	2024-03-28 10:00 - 2024-03-28 12:00	101	1001
2	2024-03-28 12:00 - 2024-03-28 13:00	102	1002
3	2024-03-29 09:00 - 2024-03-29 10:30	201	1003
4	2024-03-30 14:00 - 2024-03-30 15:00	302	1004